I Can See U: Indicator-Based Detection of Information Operations and Topic-Aware Group Analysis

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Abstract. Social media both reflects and shapes public discourse, and its openness exposes it to manipulation by information operations (IO) campaigns. Effective identification of IO accounts requires a method that provides a representative characterization of social-media activity, derives interpretable and theoretically grounded indicators that enable human-in-the-loop, and supports group analysis to reveal coordinated themes that emerge at the group level.
The proposed framework identifies IO accounts using multiple indicators of coordinated activity captured through behavioral similarity graphs constructed from co-actions such as the shared use of hashtags, mentions, named entities, and topics, and direct interactions, including reposting or mentioning other accounts. Classification performance is examined in both transductive (within-campaign) and inductive (cross-campaign) settings, followed by group-level analysis aimed at identifying meaningful coordinated inauthentic narratives. Across six campaigns, the transductive setting achieved an average ROC-AUC of 0.99, while leave-one-campaign-out inductive evaluation achieved an average ROC-AUC of 0.933. Group-level analysis shows that nearest neighbors and k-core components form compact and structurally cohesive groups with a reasonable tradeoff between topic alignment and concentration of IO accounts.

Keywords: Social Network Analysis, Information Operations, Coordination Detection, Behavioral Similarity Graph, k-core Decomposition

1 Introduction

Social media platforms are important channels for disseminating information at scale [29]. Alongside the apparent benefits, their openness creates opportunities for targeted manipulations, known as information operations (IO) [31, 28]. Documented IO campaigns have exploited social media to shape public opinion and disrupt public discourse, particularly around geopolitical events [10, 1, 32].

These campaigns are executed through multiple coordinated accounts that act collectively to amplify content related to a specific agenda [21, 16, 22]. The

involved accounts can be automated (fully or in part) or operated by humans [21]. We refer to the accounts that sustain a campaign as IO accounts.

Existing methods to detect IO accounts combine content, account metadata, and network signals [11, 15]. One of the strongest indicators of IO is coordination. It allows IO campaigns to increase the accumulated effect of their messages, but also leaves repeated, time-bounded behavioral traces, including shared retweets, URLs, hashtags, mentions, entities, and synchronized activity [24, 23, 17, 4]. Coordination-focused approaches for IO detection analyze relations among accounts to identify groups that exhibit shared suspicious behavior [24, 16, 22].

Although existing approaches demonstrate the value of coordination, the contribution of different account-relation types and interpretable structural indicators to IO detection remains unclear.

The work compares alternative account-relation types to evaluate graph-based and topic-based indicators in transductive within-campaign and inductive cross-campaign classification. Account-level predictions then serve as leads to extract structurally cohesive groups and examine their narrative focus and IO account concentration.

We address the following research questions.

- **RQ1:** Which account-relation types should be used when constructing co-action graphs for IO account classification, and what is the contribution of text-derived relations?
- **RQ2:** How accurately can IO accounts be classified using established graph-based indicators, in transductive within-campaign and inductive cross-campaign settings?
- **RQ3:** To what extent does group-level analysis of IO accounts in the co-action graphs reveal other IO accounts and their target theme?

2 Related Work

2.1 Social Network Graph Representation.

Prior work models social-media coordination through graphs connecting accounts by shared hashtags, URLs, mentions, retweets, or textual similarity [20, 16, 24]. Other approaches use temporal windows, synchronized activity, or multiple similarity networks to capture complementary coordination signals [30, 17, 16]. Mannocci et al. [19] further demonstrate the complementary value of different co-action types, highlighting the importance of relation selection and graph construction.

2.2 IO Detection

Network-based IO detection methods use graph representations to classify suspicious accounts. Luceri et al. [16] combine multiple behavioral similarity networks with centrality and node2vec features, while Gabriel et al. [11] use generalized graph signatures for cross-operation classification. More recent methods, such as

IOHunter, combine behavioral graphs, textual representations, and graph neural networks to improve generalization across IO campaigns [22]. While learned representations support effective detection, their limited structural interpretability may hinder human-in-the-loop analysis.

2.3 Group Analysis and k-core

Group-level methods examine cohesive or influential structures within social networks. Core decomposition efficiently identifies nested subgraphs according to minimum internal degree and has been used to characterize structurally embedded and influential nodes [27, 18, 14]. Related approaches such as Weighted Focal Structure Analysis identify coordinated toxic groups by combining account-level toxicity, network influence, and community structure, emphasizing the role of cohesive groups in harmful-content propagation [9]. Together, these works motivate analyzing cohesive structures not only to identify IO accounts but also to characterize the activity and objectives shared within those groups.

3 Methods and Data

The framework comprises five stages: data pre-processing and enrichment, topic clustering, co-action graph construction, author-level classification and group analysis. Graph-based and topic-based indicators represent each author, while k-core components support group-level analysis. Raw text and textual embeddings are not required for the classification.

3.1 Datasets

The experiments use the Twitter-based *Labeled Datasets for Research on Information Operations* [26, 25]. Platform-labeled IO accounts are paired with controls that used the same hashtags on the same dates, ensuring that the groups are matched by topic and time. Six datasets were chosen following IOHunter [22] selection. Table 1 summarizes the periods, account distributions, and post volumes of the six evaluated campaigns.

Table 1. Summary of the evaluated campaigns.

Campaign Name	Period	IO Accounts	Non-IO Accounts	IO Accounts %	Posts
China-1	2008–2019	699	42,120	2%	3,670,754
Cuba	2010–2020	503	30,099	2%	6,155,331
Iran-1	2010–2018	660	5,015	12%	1,322,789
Russia-1	2009–2018	3,293	31,299	10%	3,650,083
UAE	2011–2019	3,658	10,545	26%	1,707,792
Venezuela	2012–2019	611	9,745	6%	1,972,732

3.2 Text Processing and Topic Modeling

Posts are lowercased, whitespace is normalized, and URLs and mentions are removed. The preprocessing approach was manually evaluated by human reviewers, comparing translated posts with and without text cleaning, with cleaning yielding better results. Non-English posts are translated into English, while English posts are back-translated through Spanish to ensure comparable processing. Translation is performed using the distilled NLLB-200 model [6] to enable consistent named entity recognition (NER) and link posts across different languages. Named entities are extracted from the translated text using TweebankNLP/BERTweet [13] and normalized to lowercase.

Topic modeling is performed using BERTopic [12] with normalized embeddings produced by the Twitter4SSE sentence-embedding model [8]. Minimum topic sizes of 10, 128, 256, and 512 are evaluated. Models fitted to chunks of 131,072 posts are merged using a similarity threshold of 0.7. A minimum topic size of 256 was selected, as configurations with minimum topic sizes above 100 showed similar IO classification performance across all datasets, suggesting limited sensitivity to this parameter.

3.3 Co-Action Graph & Indicators

Co-Action Graph. Let P and A denote the sets of posts and authors. Each post $p \in P$ is associated with an author $a(p)$, timestamp $T(p)$, and a set of observed entities $\mathcal{E}(p)$ comprising hashtags, account mentions, named entities, and inferred topics. Where Δ denotes a time window duration, let $C_\Delta(u, v)$ denote the set of temporal co-actions between authors u and v. A co-action is included in $C_\Delta(u, v)$ when posts p_u and p_v share an entity and satisfy $|T(p_u) - T(p_v)| \leq \Delta$. Let $I(u, v)$ denote the set of direct interactions in which one author replies to, mentions, or reposts the other. The edge weight $W(u, v)$ is the total number of valid co-actions and interactions. An edge is included in the graph when $W(u, v) > 0$, while self-loops are excluded. This defines a weighted undirected graph $G = (V, E, W)$, where nodes represent authors and edges represent behavioral similarity.

Co-actions use 12-hour or three-day time windows. Similarity relations capture shared hashtags and mentions, while text-similarity relations capture shared named entities and topics. Direct interactions comprise replies, mentions, and reposts without temporal restrictions. All seven non-empty combinations of the three relation groups are evaluated.

Graph-Based Indicators. The graph-based indicators quantify an author's level of embeddedness in the co-action graph, representing behavioral similarity and coordination with other authors. The graph-based indicators comprise degree, harmonic, eigenvector, and betweenness centrality, PageRank, core number, clustering coefficient, and Louvain community. For weighted shortest-path indicators, edge distance is defined as $1/W(u, v)$, so repeated co-actions correspond to shorter distances. Harmonic and betweenness centrality are approximated using a sample of at most 400 nodes to enable efficient computation on large graphs.

Topic-Based Indicators. Topic-based indicators characterize an author's posting behavior across topics, where strong topic concentration may indicate campaign-focused behavior. Topic-based indicators include `posting_dominance`, the proportion of an author's posts assigned to their most frequent topic, and three TF-IDF-based scores. Treating inferred topics as documents and hashtags, account mentions, and named entities as terms, each author–topic score aggregates entity weights and penalizes activity across many topics:

$$\begin{aligned} S(a,t) &= \sum_{p(a,t)} q(p), \\ KeyScore(a,t) &= S(a,t) \log\left(\frac{|\mathcal{T}| + \delta}{|\mathcal{T}_a| + \delta}\right), \qquad \delta = 1, \end{aligned} \tag{1}$$

where $q(p)$ is the sum of topic-specific TF-IDF weights in post p, $\mathcal{T}$ is the set of inferred topics, and $\mathcal{T}_a$ is the set of topics in which author a was active. This score yields `key_score_sum` (sum of an author's key scores across all topics), `key_score_dominance` (the proportion of the author's top-topic score relative to the sum of scores across all topics), and `top_topic` (the topic with the highest score). Raw text, embeddings, and individual entity occurrences are not provided directly to the classifiers.

3.4 IO Account Classification

Each author is represented by the graph-based and topic-based indicators. Logistic Regression [7], Random Forest [2], and XGBoost [3] are trained without hyperparameter tuning and evaluated using ROC-AUC and PR-AUC. AUC-based metrics are selected because they are threshold-independent, allowing performance to be compared across different networks. PR-AUC is used to account for class imbalance. The tested feature sets comprise all indicators, core number alone, and all indicators excluding harmonic and betweenness centrality.

Transductive within-campaign performance is evaluated using author-level five-fold stratified cross-validation. To assess performance with limited labeled data, models are trained on stratified samples containing 1%, 5%, 10%, 25%, 50%, and 80% of the data and evaluated on held-out 20% test sets. The within-campaign evaluation is intentionally transductive: in real-world use, the full network is available at inference time, while the account IO labels remain unknown.

Cross-campaign performance is evaluated using leave-one-campaign-out classification, with five campaigns used for training and the sixth for testing. Numeric features are min–max normalized within each campaign, while community and topic identifiers are excluded because their values are not comparable across campaigns. All models use random seed 42.

3.5 Group Analysis

Group analysis evaluates whether different group definitions identify sets of accounts sharing IO status and topical focus. Six group definitions are compared: the maximal connected component in the node's maximal k-core, direct

neighbors, Louvain community, same top topic, Random-Forest-predicted IO accounts, and XGBoost-predicted IO accounts.

Two quantities characterize each group. IO account concentration is the proportion of group members labeled as IO accounts. Topic alignment is the proportion of a focal account's group that shares its dominant topic, averaged over accounts.

4 Results

4.1 Co-Action Relations and Indicator Importance

Account-Relation Configuration Table 2 reports mean within-campaign classification performance across the evaluated campaigns and classifiers. Interaction relations alone yield the lowest performance, while similarity relations achieve substantially better results. Adding text-similarity relations to similarity relations further increases ROC-AUC from 0.944 to 0.958 and PR-AUC from 0.825 to 0.844. However, the relatively modest improvement suggests that text-similarity relations may be omitted when data availability is limited or method simplicity is prioritized. Because the difference between the best and complete configurations is negligible, subsequent experiments use all indicators and relation types.

Table 2. Mean classification performance by account-relation configuration across the evaluated campaigns and classifiers.

Account relations configuration	ROC-AUC	PR-AUC
Interactions	0.873	0.669
Similarity	0.944	0.825
Similarity + interactions	0.942	0.820
Similarity + text similarity	0.958	0.844
Similarity + text similarity + interactions	0.957	0.840
Text similarity	0.956	0.843
Text similarity + interactions	0.956	0.836

Indicator Importance Across all evaluated campaigns, core number achieves the highest mean information gain (0.182), followed by Louvain community (0.171) and degree centrality (0.145).

Topic-based indicators remain informative but obtain lower scores, ranging from 0.075 for top topic to 0.085 for posting dominance. Overall, the ranking shows that structural position and group embeddedness provide the strongest signals.

The campaign-level XGBoost SHAP analysis provides a model-specific view. Figure 1 shows that feature importance varies across campaigns, highlighting the complementary contribution of multiple indicators.

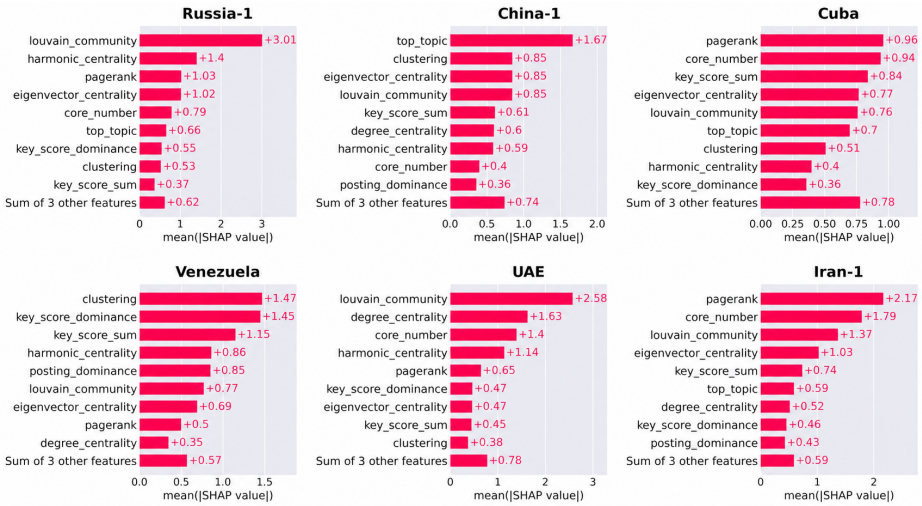


Fig. 1. XGBoost feature importance, measured by mean absolute SHAP value.

4.2 IO Account Classification Performance

Transductive Within-Campaign Classification Figure 2 reports XGBoost ROC-AUC as the fraction of labeled target-campaign data increases. With only 1% of the data, mean ROC-AUC ranges from 0.943 to 0.982 across campaigns. At 10%, all campaigns achieve at least 0.978, and performance largely plateaus beyond 25%. Classification performance when using 80% of the labeled campaign accounts is reported in Table 3. Variability is highest at the smallest training fractions and decreases as more labeled accounts become available. Table 4 shows that ICSU (I Can See You) achieves the highest average Macro-F1 (0.975), outperforming IOHunter (0.964), node2vec+RF (0.878), and SBERT (0.857).

Inductive Cross-Campaign Classification Table 3 reports inductive cross-campaign performance. Random Forest and XGBoost achieve mean ROC-AUC values of 0.933 and 0.926 and mean PR-AUC values of 0.723 and 0.679, respectively. Transfer is strongest for Iran-1, UAE, and Cuba. Random Forest provides the stronger mean cross-campaign performance. Table 4 shows that ICSU achieves a higher average inductive Macro-F1 (0.804) than IOHunter (0.793).

4.3 Group Analysis

Figure 3 compares IO account concentration and topic alignment across the six group definitions, using all accounts and XGBoost-predicted IO accounts. Restricting group construction to XGBoost-classified IO accounts markedly changes the structural groups, significantly increasing the mean IO concentration.

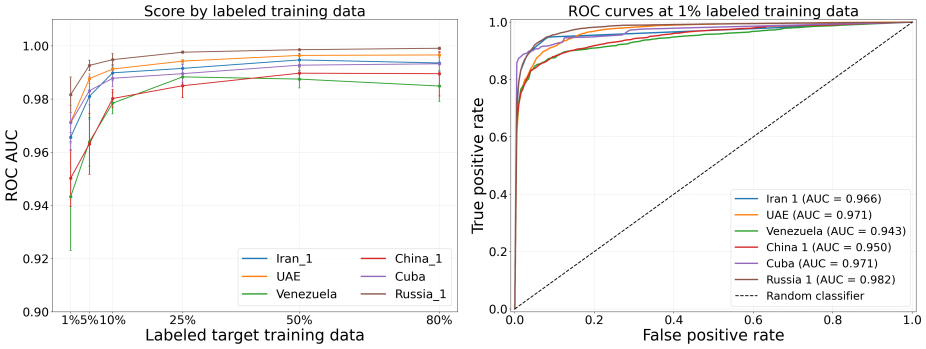


Fig. 2. Transductive within-campaign XGBoost ROC-AUC across labeled target-campaign training fractions. Points are means and error bars are one standard deviation across five stratified splits.

Table 3. Classification performance in transductive and inductive settings, with random forest (RF) and XGBoost (XGB). Transductive use 80% of the labeled accounts for training and are averaged over five stratified splits. Inductive values use leave-one-campaign-out evaluation.

	Transductive (80%)				Inductive			
Target campaign	RF ROC	XGB ROC	RF PR	XGB PR	RF ROC	XGB ROC	RF PR	XGB PR
Iran-1	0.991	0.994	0.983	0.985	0.989	0.974	0.963	0.899
UAE	0.995	0.997	0.992	0.994	0.976	0.979	0.943	0.949
Venezuela	0.984	0.985	0.964	0.966	0.889	0.896	0.649	0.658
China-1	0.986	0.990	0.940	0.934	0.822	0.809	0.072	0.047
Cuba	0.985	0.993	0.966	0.968	0.992	0.965	0.920	0.730
Russia-1	0.999	0.999	0.994	0.995	0.929	0.935	0.794	0.789
Mean	0.990	0.993	0.973	0.973	0.933	0.926	0.723	0.679

The analysis showed that k-core groups consistently outperformed Louvain groups and provided a better balance between IO account concentration and topic alignment. Neighborhood groups also performed well, but their performance was more strongly associated with IO concentration than topic alignment.

Figure 4 provides a structural view of the co-action graph. The same layout is colored by core number in the left panel and by ground-truth IO labels in the right panel. High core number nodes visibly overlap with concentrations of IO accounts. This correspondence is consistent with the high information gain of core number and the SHAP analyses. Maximal k-core components consequently provide an interpretable group definition: they identify mutually embedded sets of authors rather than isolated high-scoring accounts.

Case Study A connected component of the maximum k-core in the Iran 1 account graph was selected, comprising 312 accounts, all labeled as IO accounts in

Table 4. Macro-F1 comparison of ICSU using a Random Forest (RF) classifier against node2vec+RF, SBERT, and IOHunter [22]. The transductive ICSU results used 80% of the data for training, while the baseline results were taken from [22], which used a 60/20/20 train/validation/test split. For China, Iran, and Russia, ICSU was evaluated on a single campaign per country, whereas the baseline methods were evaluated on all available campaigns within each country. Δ denotes ICSU minus IOHunter.

	Transductive					Inductive		
Campaign	ICSU	node2vec +RF	SBert	IOHunter	Δ	ICSU	IOHunter	Δ
China / China 1	0.948	0.839	0.759	0.929	+0.019	0.513	0.581	-0.069
Cuba	0.984	0.915	0.904	0.989	-0.005	0.923	0.899	+0.024
Iran / Iran 1	0.982	0.805	0.852	0.966	+0.015	0.938	0.728	+0.210
Russia / Russia 1	0.981	0.834	0.840	0.923	+0.059	0.864	0.798	+0.066
UAE	0.981	0.970	0.862	0.988	-0.007	0.897	0.839	+0.057
Venezuela	0.973	0.903	0.927	0.991	-0.018	0.688	0.910	-0.222
Average	**0.975**	**0.878**	**0.857**	**0.964**	**+0.010**	**0.804**	**0.793**	**+0.011**

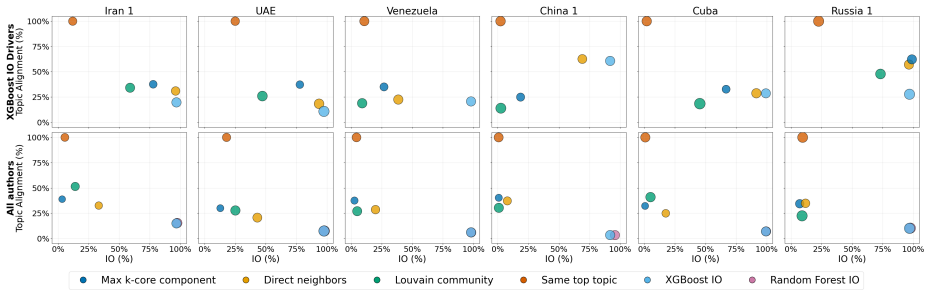


Fig. 3. IO account concentration and topic alignment by group definition, aggregated across all accounts (bottom) and XGBoost-predicted IO accounts (top) in each campaign. Each point is the mean for one group type.

the ground truth. Peak activity was identified using non-overlapping 15-minute intervals. Each interval was scored by summing the number of distinct account pairs sharing each hashtag. A pair sharing multiple hashtags contributed once for each shared hashtag. The highest-scoring interval, 13:00–13:15 on December 2, 2017, contained 244 posts from 15 active accounts, including 242 re-posts and two original posts, and involved 30 shared hashtags. Re-posting was performed by 13 accounts, with no account reposting the same source more than once.

The interval contained 21 unique posts: two original and 19 reposted source posts. Of the source posts, 16 concerned U.S. politics, two celebrated ISIS victories, and one contained a hate message. The posts mainly criticized President Donald Trump through calls for impeachment, opposition to tax legislation (#taxcuts, #taxscambill, #taxreform), and sexual misconduct allegations against Republican Senate candidate Roy Moore (#roymoorechildmolester). Each hashtag appeared 26 times. The activity preceded the December 12, 2017, Alabama Senate election, reflecting contemporary U.S. political controversies.

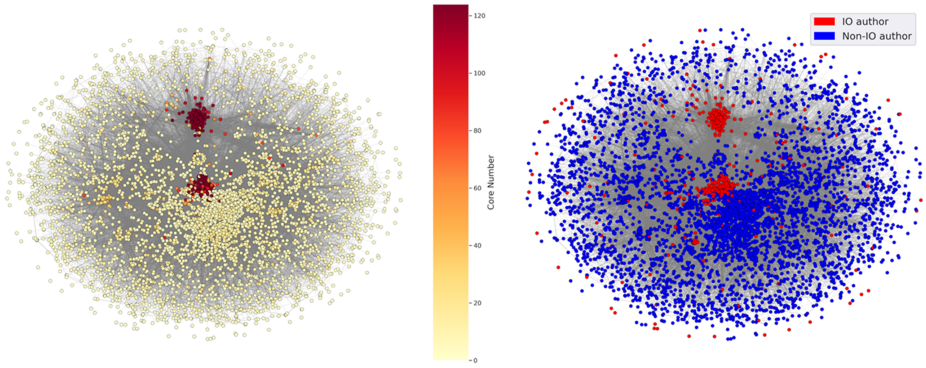


Fig. 4. Co-action graph shown with a spring layout colored by core number (left) and IO ground truth (right) for the Iran-1 campaign.

5 Conclusions

The results show that an indicator-based method with behavioral similarity co-action graphs effectively distinguishes IO accounts from legitimate accounts. Shared behavioral and text-derived co-actions are more informative than direct interactions, while multiple graph-based indicators provide complementary classification signals.

This work systematically evaluates how different types of co-action relations affect IO account detection in behavioral similarity graphs. The proposed framework uses interpretable indicators to support IO detection and human-in-the-loop analysis in both transductive within-campaign and inductive cross-campaign settings. Furthermore, it integrates account-level classification with group-level analysis, enabling the investigation of coordinated structures and their shared narrative focus.

The study is limited by the age of the data and by the procedure used to select controls based on shared hashtags and posting dates. These controls may not capture the full diversity of non-IO behavior, potentially limiting real-world deployment. Translation, topic inference, and named-entity extraction may also introduce processing errors, while group interpretation remains sensitive to campaign-specific characteristics.

Future work should evaluate the framework on more recent datasets with alternative control-sampling strategies and incorporate culture-aware analysis of the detected groups.

Acknowledgments

The authors thank Prof. Doron Havazelet for his valuable feedback throughout the development of this research.

Code Availability

The code used for the analysis and figure generation is available on GitHub [5].

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